

Practice set 2: Discrete Mathematics (CSE121)

Instruction. This document is solely for practice. Do not submit the answers. Also, the answer keys to these questions will not be provided.

1 Problems

Question 1.1. *What will be the conclusion of the following statements?*

1. *If I wake up early, then I will exercise. If I exercise, then I will feel energetic. I woke up early.*
2. *If a number is divisible by 4, then it is even. 12 is divisible by 4.*
3. *If I study hard, I will pass the exam. I studied hard.*
4. *If a function is differentiable, then it is continuous. If a function is continuous, then it is bounded on a closed interval. $f(x)$ is differentiable.*
5. *If a shape is a square, then it is a rectangle. If a shape is a rectangle, then it is a quadrilateral. This shape is a square.*
6. *If a car has fuel, then it will start. If it starts, then it can move. This car has fuel.*
7. *If a seed is watered, then it will sprout. If it sprouts, then it will grow into a plant. If it grows into a plant, then it will produce flowers. The seed is watered.*
8. *If a student attends lectures, then they learn concepts. If they learn concepts, then they solve problems. If they solve problems, then they pass the exam. Raj attends lectures.*
9. *If a number is divisible by 12, then it is divisible by 6. If it is divisible by 6, then it is divisible by 3. If it is divisible by 3, then it is not prime. If it is not prime, then it is composite. 60 is divisible by 12.*
10. *If a person is kind, then they help others. If they help others, then they gain respect. If they gain respect, then they are remembered. If they are remembered, then they achieve immortality in memory. Ram is kind.*

Question 1.2. *Give direct proofs for the following using the rules for integers mentioned in class.*

1. *Prove that the sum of two even integers is even.*
2. *Prove that the sum of two odd integers is even.*

3. Prove that the product of two odd integers is odd.
4. Prove that the product of an even integer and a prime number is always even.
5. Prove that if n is even, then n^2 is even.
6. Prove that if a and b are integers and both are even, then $4 \mid ab$.
7. Prove that if $6 \mid n$, then $2 \mid n$ and $3 \mid n$.
8. Prove that if $12 \mid n$, then $36 \mid n^2$.
9. Prove that if $4 \mid a$ and $6 \mid b$, then $24 \mid ab$.
10. Prove that for any integer n , $n^2 + n$ is even.
11. Prove that if n is an integer, then $n^3 - n$ is an even number. (Hint: Consider two cases, when n is even and when it is odd).
12. Prove that the product of two consecutive integers is even.
13. Prove that the product of two consecutive integers is divisible by 2, and the product of three consecutive integers is divisible by 6.
14. Prove that for any integer n , $5 \mid n^5 - n$. (Hint: Consider two cases, when n is even and when it is odd).
15. Prove that if a and b are odd integers, then $4 \mid a^2 - b^2$.
16. Prove that for every integer n , $n^2 - n$ is always even.
17. Prove that the sum of two consecutive odd integers is divisible by 4.
18. Prove that for every integer n , $n^2 + 3n$ is even.
19. If n is an odd integer, then prove that $8 \mid n^2 - 1$.
20. Prove that the square of every integer is of the form $4k$ or $4k + 1$ for some integer k .

Question 1.3. Prove the following by contraposition.

1. Prove that if the square of an integer is even, then the integer itself is even.
2. Prove that if the product of two integers is odd, then both integers are odd.
3. Prove that if $4 \mid n^2$, then $2 \mid n$.
4. Prove that if an integer is not divisible by 2, then its square is not divisible by 4.
5. Let m, n be integers. If mn is even, then at least one of them is even.
6. Prove that if $a^2 < b^2$ for positive integers a, b , then $a < b$.
7. Prove that if n is prime, then $n > 1$.
8. Prove that if an integer n satisfies $n^2 < n$, then n must be negative.
9. Prove that if an integer n satisfies $n^2 > 2n$, then $n > 2$ or $n < 0$.
10. Let a, b be integers. If $a + b$ is odd, then one of a is odd and the other is even.